Enrollment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Computer Application
MCA | Sem-3 | Winter-2025

Course Code	:	2305CS302	**Date**	:	14-10-2025
Course Name	:	Introduction to Advanced Technologies	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	What is cloud computing? Why it is important?	**4**
	(B)	Describe the characteristics of cloud computing.	**3**
		OR	
		Explain advantages of cloud computing.	
	(C)	Explain the types of cloud.	**7**
		OR	
		Describe cloud service models.	
Q.2	**(A)**	Compare AR and VR.	**4**
	(B)	List out challenges of AR and VR.	**3**
		OR	
		Describe functionalities of AR.	
	(C)	Explain the visualization techniques of AR and VR.	**7**
		OR	
		List out the applications of AR and VR. Explain any two applications in details.	
Q.3	**(A)**	Describe Artificial Neural Network?	**4**
	(B)	Define Artificial Intelligence and Machine Learning.	**3**
		OR	
		Lists applications of Machine Learning.	
	(C)	Explain Machine Learning life cycle in details.	**7**
		OR	
		Describe classification of machine learning techniques.	

Q.4	**(A)**	What is Data Science? Explain its importances.	**4**
	(B)	Write steps of Exploratory Data Analysis (EDA)	**3**
		OR	
		Write short note on Data visualization.	
	(C)	What are Data Science components? Explain the steps for creating pipeline in data science.	**7**
		OR	
		Give the list of libraries used for data science. Explain the use of NumPy and Pandas library with at least two functions.	
Q.5	**(A)**	Explain different components of blockchain.	**4**
	(B)	Give understanding of blockchain transaction step by step.	**3**
		OR	
		Define blockchain technology and cryptocurrency.	
	(C)	List various consensus algorithm. Give fundamental working of consensus any two algorithms.	**7**
		OR	
		Explain types of blockchain in detail.	

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